7.6. ABSORPTION OF NEUTRONS

A beam of collimated neutrons is attenuated in a thin absorber through scattering and absorption processes in a similar manner to the attenuation of γ-rays; these processes are described in previous sections. In a thick absorber the neutrons are slowed from incident energy at the absorber face to thermal energies if the absorber is thick enough. The ultimate fate of the neutron is capture in an absorber nucleus. Because of the spread in neutron energy and the energy dependency of the capture cross-sections, no simple relation can be given for the attenuation of the neutron beam (cf. next section).

7.7. RADIATION SHIELDING

The absorption properties of nuclear radiation in different materials must be known in order to design shielding to avoid unwanted radiation effects on the surroundings by nuclear radiation sources.

For charged particles the shielding is usually slightly thicker than that required for the maximum range of the particles in the material. Absorption thicknesses of 0.2 mm are adequate to completely absorb the particles from α-decay. By contrast 15 mm of materials of low Z such as water, plastic, etc., are required for absorption of β-radiation with energies up to 3 MeV. Radiation shielding constructed from materials of higher atomic number require correspondingly thinner thicknesses. The data in Table 7.2 and the curves in Figs. 7.6 and 7.13 provide information on the thickness of absorber material required for the energy of various types of radiation.

Since γ-rays and neutrons have no definite range but exhibit a logarithmic relation between thickness and intensity, only a partial reduction of the radiation can be obtained. By combining Eqns (7.7) and (7.8), Eqn (7.27) is obtained

$$\varphi = \left(n\,A/\left(4\pi\,r^2\right)\right)\mathrm{e}^{-\mu x} \tag{7.27}$$

It is seen that the intensity from a point source of radiation can be decreased by increasing either the distance from the source r or the thickness x of the absorber. Alternately, an absorber with a higher absorption coefficient, μ, can be chosen to reduce the thickness required.

Equation (7.27) is valid only for point sources with ideal geometry, i.e. no back or multiple scattering, etc. For γ-radiation the thicker the absorber the higher is the percentage of radiation which may be scattered backwards through secondary (mainly Compton) scattering. The effect of geometry and absorber thickness can be taken into account by including a constant B in the absorption equation:

$$\varphi = B\,\varphi_0\,\mathrm{e}^{-\mu x} \tag{7.28}$$

The "*dose build-up*" *factor* B not only takes into account multiple Compton and Rayleigh scattering but also includes correction for positron formation at high γ-energies